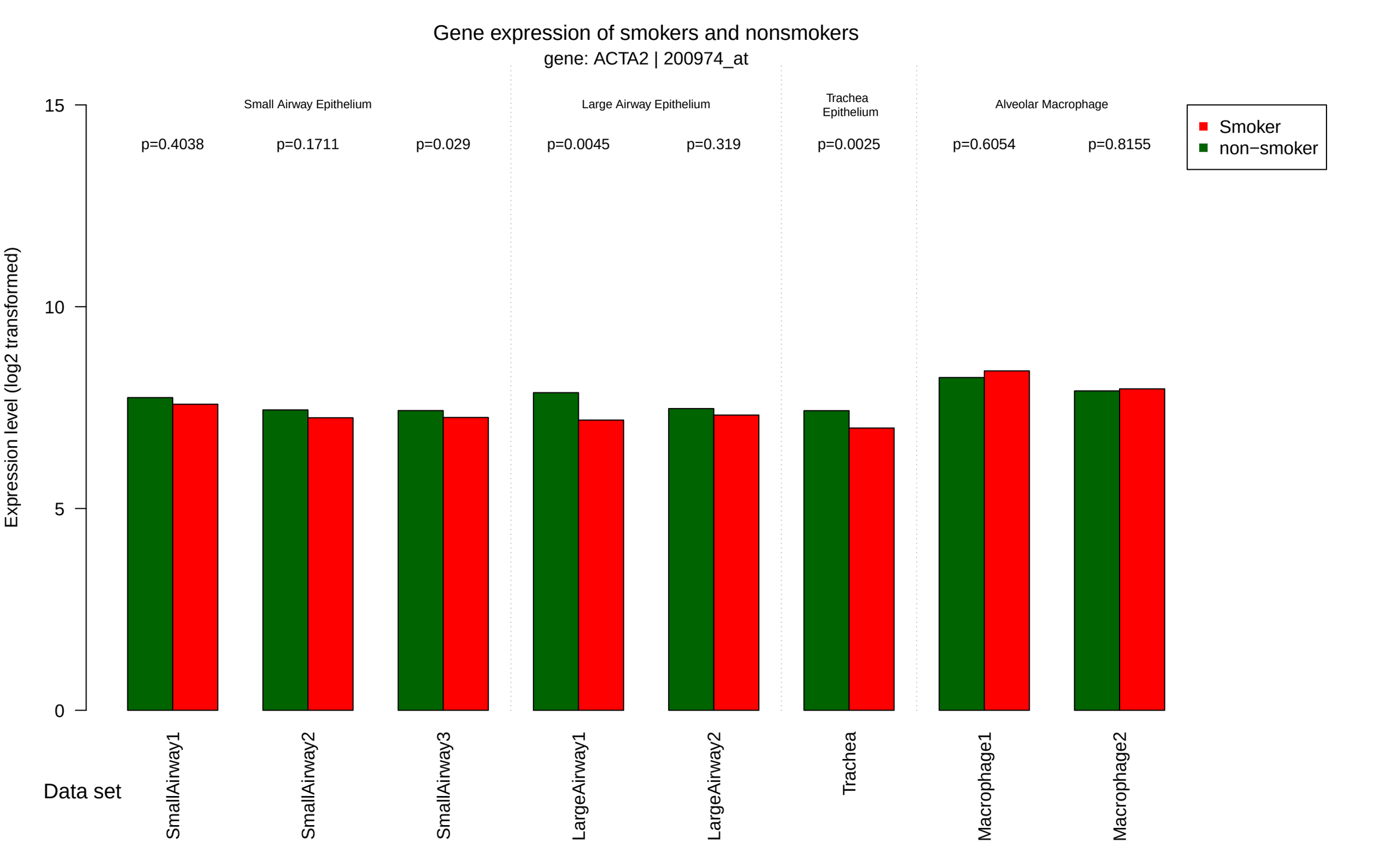
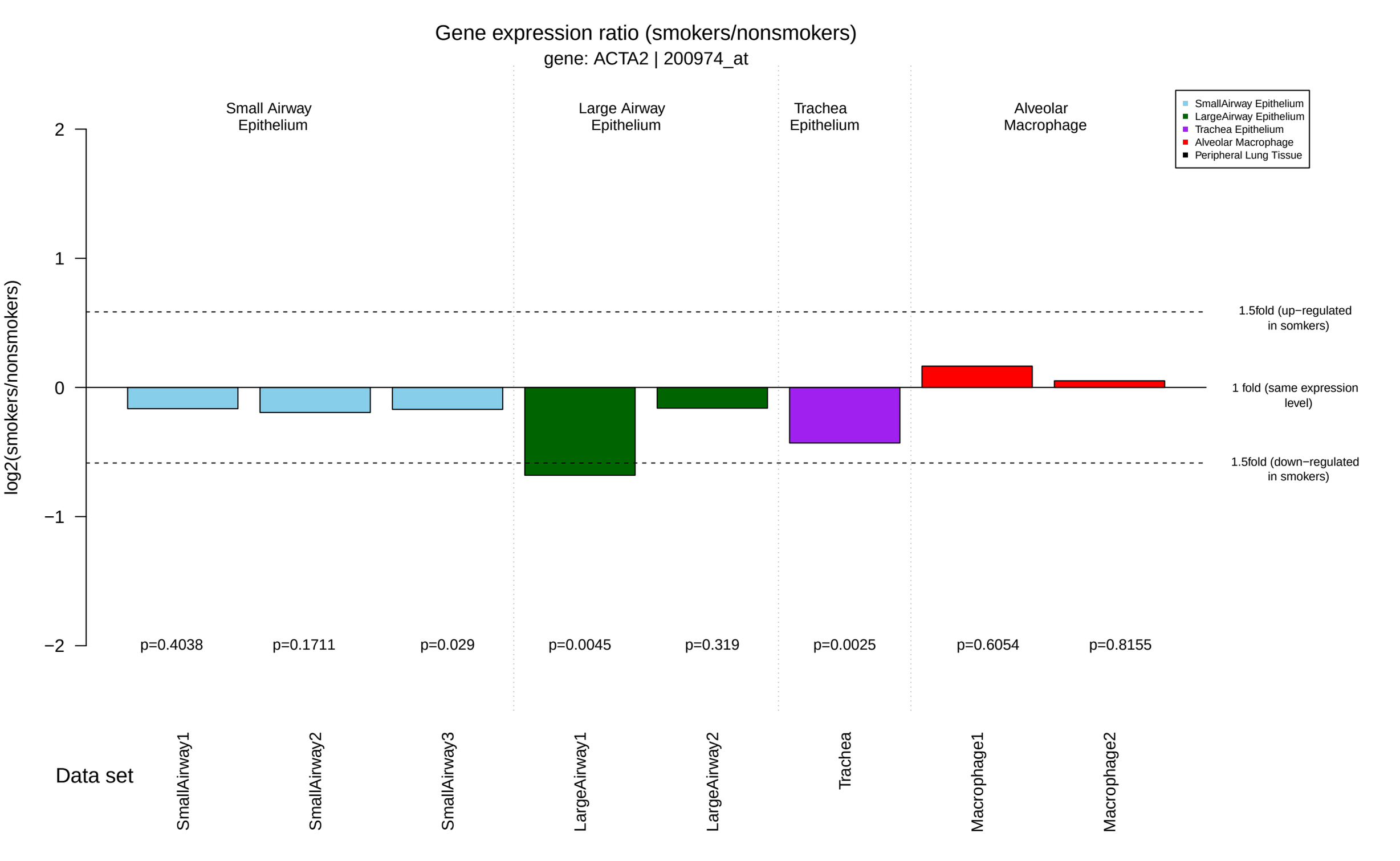
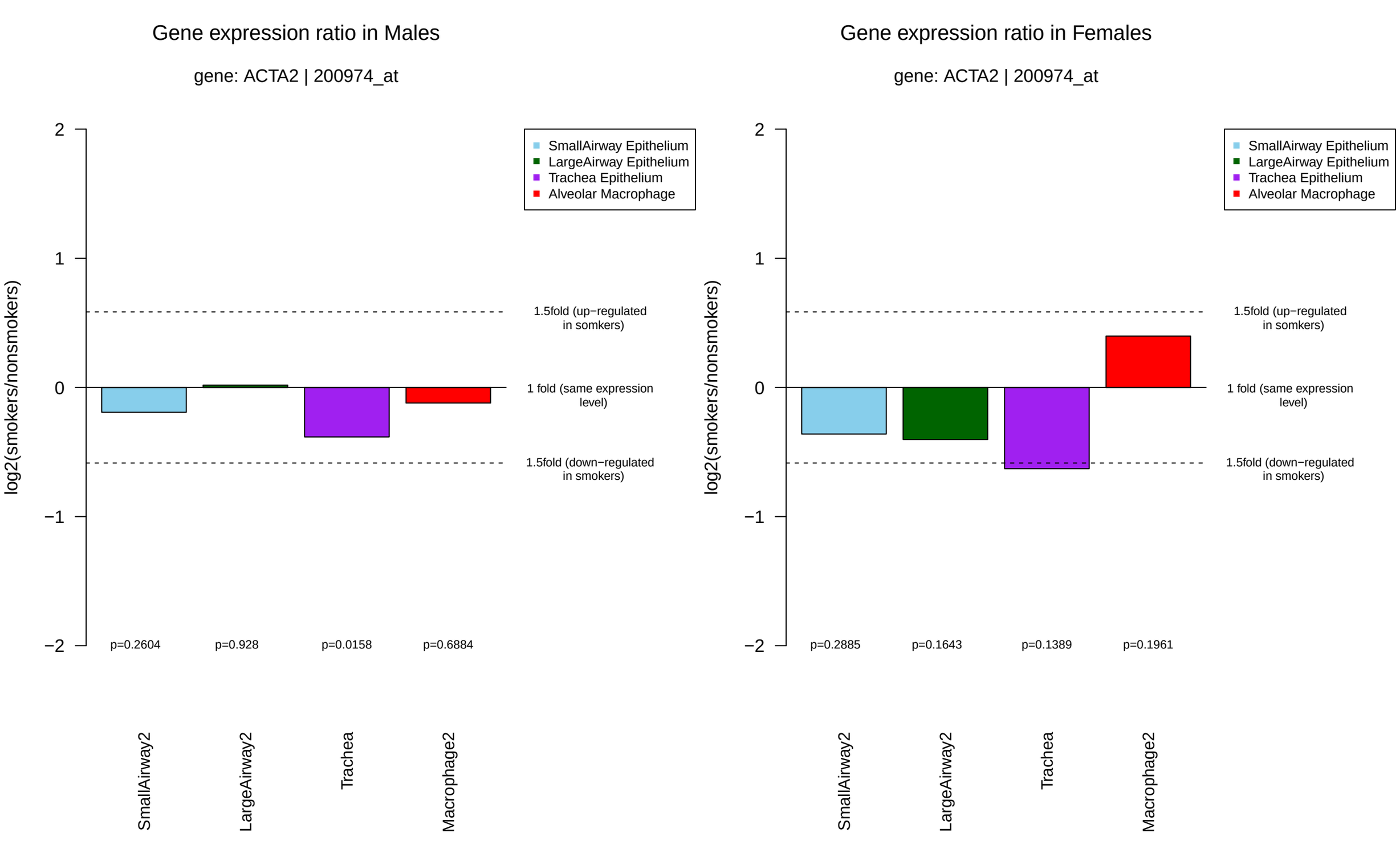
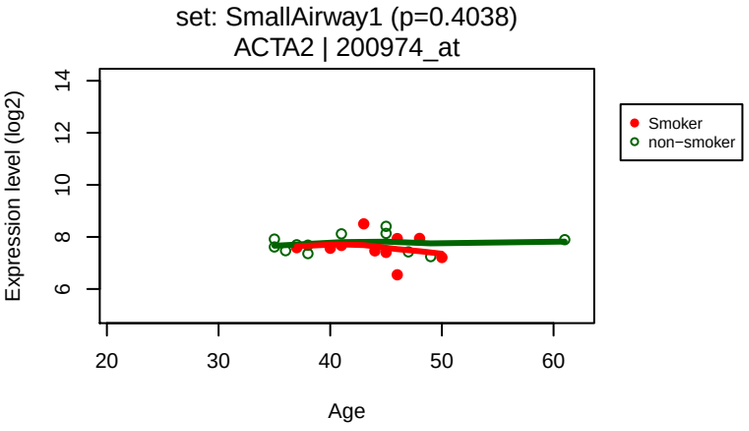
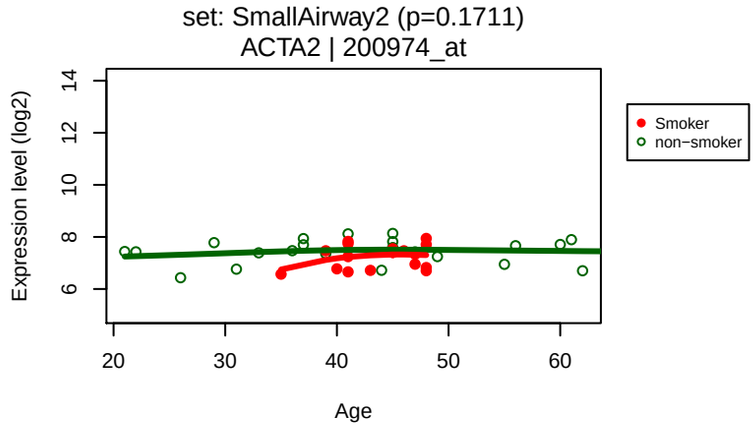
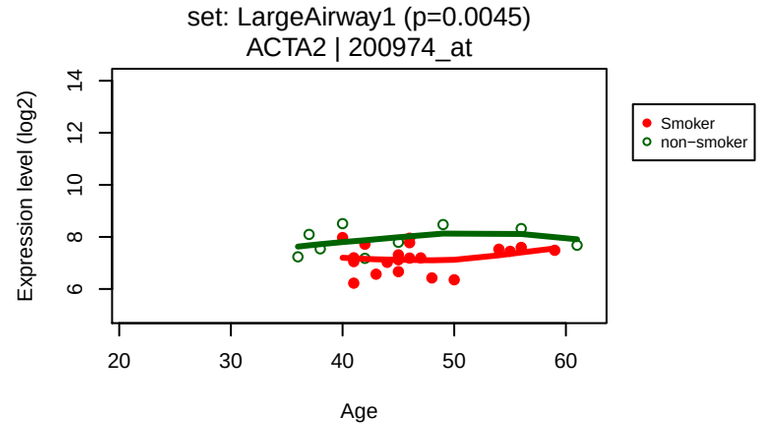

Gene expression of smokers and nonsmokers
gene: ACTA2 | 200974_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4038
p=0.1711
p=0.029
p=0.0045
p=0.319
p=0.0025
p=0.6054
p=0.8155
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

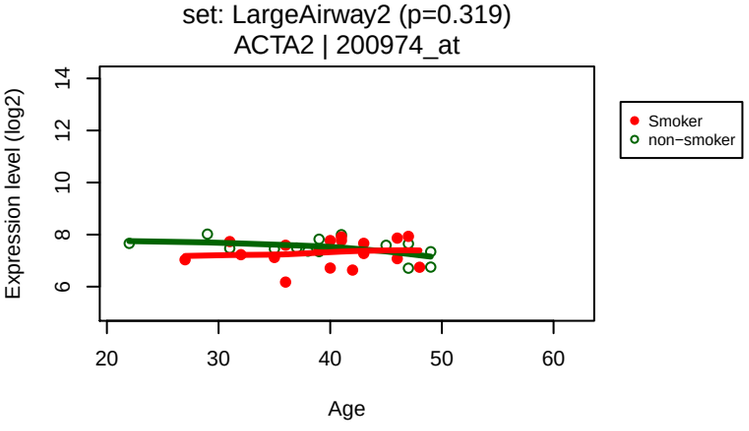

Gene expression ratio (smokers/nonsmokers)
gene: ACTA2 | 200974_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4038
p=0.1711
p=0.029
p=0.0045
p=0.319
p=0.0025
p=0.6054
p=0.8155
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

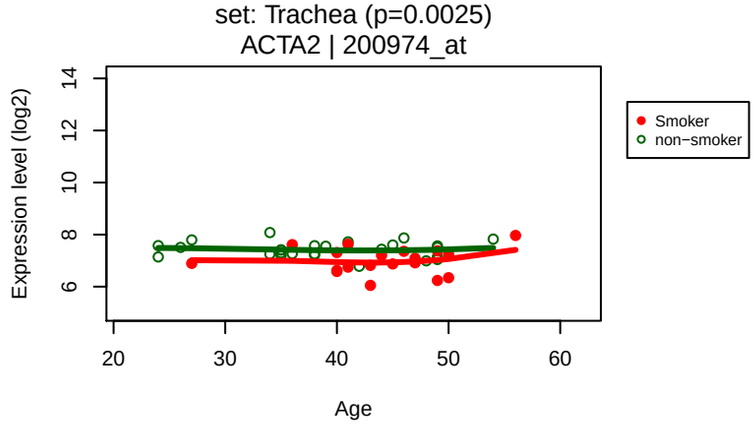

Gene expression ratio in Males
gene: ACTA2 | 200974_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2604
p=0.928
p=0.0158
p=0.6884
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTA2 | 200974_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2885
p=0.1643
p=0.1389
p=0.1961
SmallAirway2
LargeAirway2
Trachea
Macrophage2

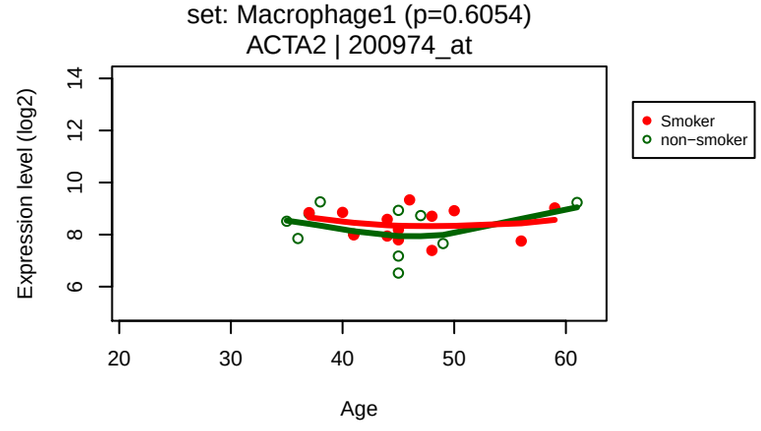

set: SmallAirway1 (p=0.4038)
ACTA2 | 200974_at
Expression level (log2)
Age
Smoker
non−smoker

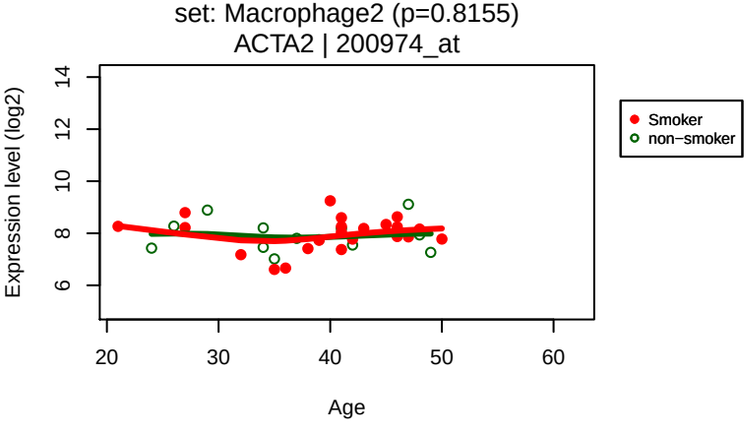

set: SmallAirway2 (p=0.1711)
ACTA2 | 200974_at
Expression level (log2)
Age
Smoker
non−smoker

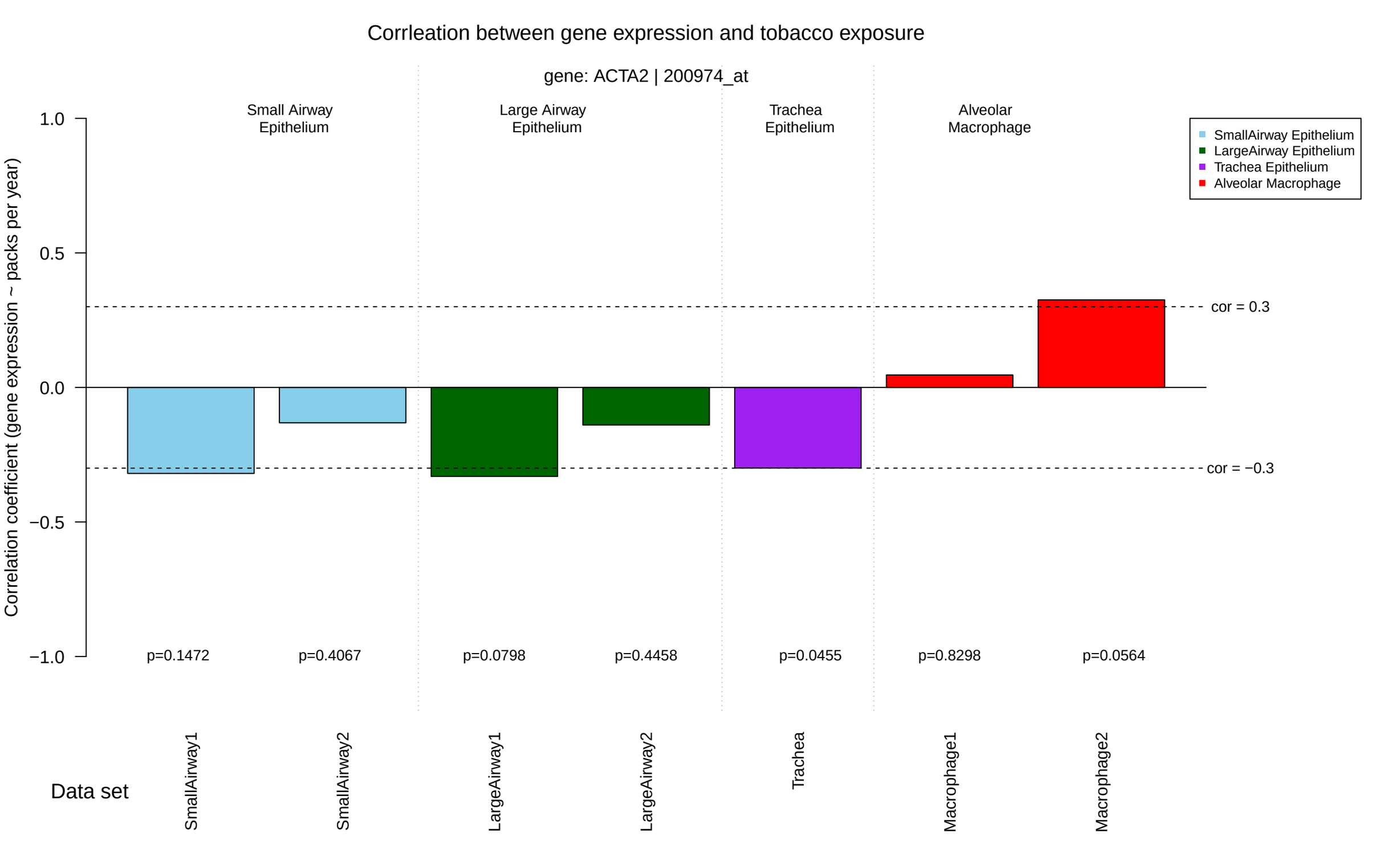

set: LargeAirway1 (p=0.0045)
ACTA2 | 200974_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.319)
ACTA2 | 200974_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0025)
ACTA2 | 200974_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6054)
ACTA2 | 200974_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8155)
ACTA2 | 200974_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTA2 | 200974_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1472
p=0.4067
p=0.0798
p=0.4458
p=0.0455
p=0.8298
p=0.0564
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2